

Day 4 lab: quantization

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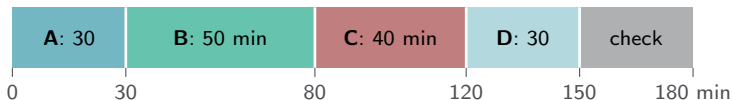
- ▶ **Goal:** your group has a measured comparison of a `float32` model and an `int8` model: size, accuracy, and latency, on the laptop and on the XIAOML Kit.
- ▶ **Board:** the XIAOML Kit. Only Part D needs the kit.
- ▶ **Deliverable:** the file `report.md` with the comparison table of Part D and the Decision Log.

From the lecture to the lab

The lecture gave the mapping, the two methods, and the search for the layer that loses accuracy. In the lab, you write the mapping, you quantize a model with the two methods, and you measure the float model and the `int8` model on the board.

Key point: Each `int8` number needs the float number next to it, from the same test set and the same board.

Plan of the three hours



Part	Work	Time
A	By hand: quantize tensors in NumPy, measure the rounding error and the clipping error	30 min
B	Post-training quantization: quantize a small CNN, change the calibration set	50 min
C	Quantization-aware training: train the same CNN with simulated quantization	40 min
D	On the board: the float model and the int8 model of Day 3	30 min
Check by the instructor		30 min

Write each result in `report.md` when you get it.

Hardware for each group

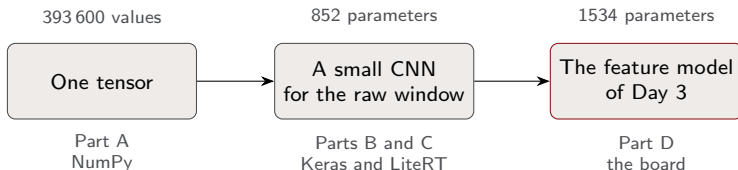
- ▶ XIAOML Kit with the Arduino firmware. You need it in Part D only.
- ▶ USB-C data cable.
- ▶ Laptop with your Day 2 dataset in `Labs/day02/data/`.

Software

- ▶ Arduino IDE 2 with the esp32 core 3.3.12.
- ▶ The library `Chirale_TensorFlowLite 2.0.0` of Day 3.
- ▶ The Python packages of Day 3: `tensorflow-cpu` and `ai-edge-litert`.
- ▶ No new package and no new library.

Lab files	<code>Labs/day04/</code> . Run the commands from this folder.
Notebook	<code>../.../venv/bin/jupyter lab quantization.ipynb</code>
Sketch	<code>sketches/motion_quant/motion_quant.ino</code>

Three objects, one method



- ▶ **Part A:** you write the mapping for one tensor: the acceleration values of your training windows.
- ▶ **Parts B and C:** the converter and the training apply the same mapping to each tensor of a CNN. The CNN takes the raw window in g , so the range of its input depends on the motion.
- ▶ **Part D:** the model of Day 3 goes to the board two times: in `float32` and in `int8`.

Source: The number of values is the number for the fallback dataset. Your dataset gives a different number.

Part A: quantize by hand (30 min)

Run sections 0 and 1 of the notebook. Then write three functions in section 2.

Task	You write	The notebook checks
1	<code>scale_zero_point</code> : the scale and the zero point of a range	The range $[-1.00, 1.55]$ gives $S = 0.01$ and $Z = -28$
2	<code>quantize</code> and <code>dequantize</code> : one line for each function	The five numbers of the lecture exercise give $-28, 22, -61, 95$, and 127
3	Your prediction: the range with the smallest error	Not checked. You compare it with the result.

$$S = \frac{r_{\max} - r_{\min}}{255} \quad Z = \text{round}\left(-128 - \frac{r_{\min}}{S}\right) \quad q = \text{clamp}\left(\text{round}\left(\frac{r}{S}\right) + Z\right)$$

- Use `np.round` and `np.clip`. The clamp limits the result to the range from -128 to 127 . The way back is $\hat{r} = S(q - Z)$.

Section 3: the rounding error

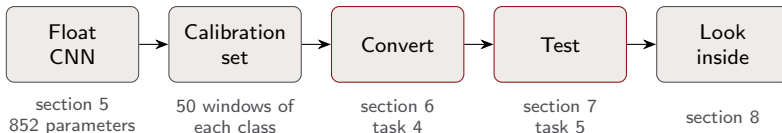
- ▶ The code quantizes all acceleration values of your training windows.
- ▶ **You see:** the largest error and the typical error.
- ▶ Check the two rules of the lecture: $S/2$ and $S/\sqrt{12}$.
- ▶ Then the same range with 7, 6, 4, and 2 bits.
- ▶ **You record:** the scale, the zero point, the two errors, and the answers to the two questions of section 4.

Section 4: the clipping error

- ▶ Four ranges for the same tensor: the minimum and the maximum, and the ranges that hold 99.9, 99, and 90 percent of the values.
- ▶ **You see:** the scale, the clipped values, and the RMS error, with 8 bits and with 4 bits.

Prediction (task 3). Which of the four ranges gives the smallest RMS error with 8 bits? Which range with 4 bits? Write the prediction before you run the experiment.

Part B: post-training quantization (50 min)



Task	You write	The notebook checks
4	<code>convert_int8</code> : the generator of the calibration inputs, and the five settings of the converter	The input tensor of the model has the type <code>int8</code>
5	<code>quantize_input</code> : the <code>int8</code> input of the model, and the number of clipped values	The numbers of the lecture exercise, with two clipped values

- The generator gives one input for each `yield`: a list with one array of the shape `(1, 100, 3)`.

Section 7 prints one row for each model. The accuracy and the clipped input values are in percent:

model	bytes	accuracy	idle	terrest	lift	maritim	clipped
float32	...	...	...	...	...	...	0.00
int8, all classes	...	...	...	...	...	...	...

- ▶ **You record:** the two rows, the largest difference between the two outputs, and the next step by the rule of the lecture.
- ▶ The file does not become four times smaller. Find the reason in the output of section 6.

Section 8 prints each tensor of the int8 model that has a scale. Answer three questions in the report:

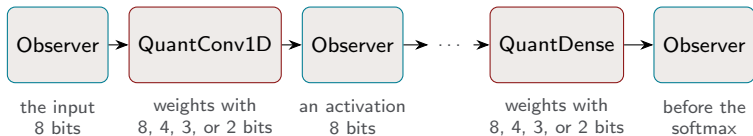
1. Which scale and which zero point has the input tensor? Compare them with the result of your function for the range of the calibration set.
2. How many scales has the weight tensor of each convolution, and which zero point?
3. Which type have the biases?

The conversion gives no error when the calibration set is wrong. Section 9 makes this error on purpose.

Experiment	Calibration set	You record
1	50 windows of one class only. One conversion for each class.	The accuracy, the accuracy of each class, and the clipped input values
2	All classes, with 200, 20, and 4 windows	The accuracy and the clipped input values

- ▶ The table of section 1 gives the smallest value and the largest value of each class. Use it for your prediction.
- ▶ The column `clipped` needs no labels. It shows the problem also for inputs with no known class.

Prediction (task 6). Which calibration class gives the lowest accuracy? Is there a class that gives no loss? Which is the smallest set of all classes with no loss: 4, 20, or 200 windows?



- ▶ The notebook builds the CNN again with layers that quantize and dequantize in the forward pass. All layers use your function `fake_quant`.
- ▶ **Task 7.** Replace the last line of `fake_quant` with the straight-through estimator. Use `ops.stop_gradient`.
- ▶ The notebook checks two facts: the forward value is the quantized value, and the gradient is 1 for each value.
- ▶ **You record:** the accuracy with simulated quantization with 8 bits. It must be near the accuracy of the `int8` LiteRT model of Part B.

Key point: With no straight-through estimator, the gradient is `None`, and the training cannot start.

Section 11 trains the model four times. The activations keep 8 bits:

weight bits	values	no new training	quantization-aware training
8	255	...%	...%
4	15	...%	...%
3	7	...%	...%
2	3	...%	...%

- ▶ “No new training” is post-training quantization. LiteRT has no format below 8 bits: the rows for 4, 3, and 2 bits are a simulation only.
- ▶ **Section 12:** one layer has weights with 4 bits or with 2 bits. All other values stay in float. **You record:** the loss of each layer, and the layer that you keep at 8 bits.
- ▶ **Section 13:** the model that trained with 8 bits becomes a real int8 file. **You record:** the three rows of the table.

Prediction (task 8). Which accuracy has the model with weights of 2 bits and no new training? Does quantization-aware training repair the loss?

Part D: on the board (30 min)

Step	Work	Time
1	Run sections 14 and 15 of the notebook. They train the feature model of Day 3 and convert it two times.	5 min
2	Write your prediction in the report.	2 min
3	Upload the sketch with the float model. Record the numbers.	6 min
4	Complete tasks D1 and D2. Upload the sketch with the int8 model. Record the numbers.	8 min
5	Make the four motions. After the test set, the sketch classifies the motion of the kit, as on Day 3.	4 min
6	Complete the comparison table. Write the Decision Log.	5 min

- ▶ The notebook writes four files into `sketches/motion_quant/`: `model_float.h`, `model_int8.h`, `model_settings.h`, and `test_set.h`.
- ▶ `test_set.h` holds 80 test windows, with the class that each model gives on the laptop.

Open the sketch `motion_quant.ino`. Set Tools > PSRAM > Disabled. The line `#define MODEL_INT8 0` selects the float model. With the value 1, it selects the int8 model.

Task	You write, in the function <code>classify()</code>
------	--

- | | |
|----|---|
| D1 | Quantize the input with the scale and the zero point of <code>input->params</code> . Calculate <code>lroundf(x / scale) + zero_point</code> . Limit the result to the range from -128 to 127 . Write it into <code>input->data.int8[i]</code> . Count the clipped values. |
| D2 | Dequantize the output with the scale and the zero point of <code>result->params</code> . Write <code>scale * (result->data.int8[k] - zero_point)</code> into <code>output[k]</code> . |
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- ▶ The float model needs no change. Upload it first.
- ▶ When task D1 is complete, remove the line `task_d1_complete = false;`. Do the same for task D2.
- ▶ Before that, the sketch with the int8 model prints that the tasks are not complete, and it stops.

The Serial Monitor at 115200 baud shows these lines for the int8 model:

```
Model:                int8
Model size (flash):   ... bytes
Sketch size (flash):  ... bytes
Arena size:           4096 bytes
Arena used:           ... bytes
Input tensor:  type INT8, 63 bytes, scale ..., zero point ...
Output tensor:  type INT8, 4 bytes, scale 0.003906, zero point -128
Test set:       80 windows
Test set, correct:  ... of 80 (... percent)
Test set, same class as on the laptop: ... of 80
Test set, clipped input values: ... of 5040 (laptop: ...)
Test set, invoke_us:  median ..., largest ...
```

- ▶ The sketch runs the 80 test windows on the board. The line `same class as on the laptop` must show 80 of 80. If it does not, check tasks D1 and D2.
- ▶ The float model prints the same lines, with the type `FLOAT32` and with no line for the clipped values.

Source: The output shows the form of the lines. Nobody measured the values on the kit. The output scale is the scale of a softmax in int8: 1/256.

Number	Where you read it, for each model
File size, test accuracy, peak of the activations	Section 14 of the notebook
Flash	Sketch uses ... bytes of the build
Arena	Arena used of the Serial Monitor
Accuracy on the board	Test set, correct
Agreement with the laptop	Test set, same class as on the laptop
Latency	Test set, <code>invoke_us</code> : the median and the largest value

Write the two values and the change of each number. Then compare each result with your prediction.

Prediction. How does the `int8` model change the flash use, the arena, and the time of `Invoke()`? The ESP32-S3 has a circuit for float arithmetic, and the library has no optimized kernels for this chip.

The instructor checks this at the end of the lab:

- ▶ ✓ The notebook prints complete for tasks 1, 2, 4, 5, and 7.
- ▶ ✓ The report has the four predictions: task 3, task 6, task 8, and the prediction of Part D. You wrote them before the measurement.
- ▶ ✓ The report has the table of the calibration experiment, with the accuracy of each class and the clipped input values.
- ▶ ✓ The Serial Monitor shows Test set, same class as on the laptop: 80 of 80 for the int8 model.
- ▶ ✓ The comparison table has a float32 column and an int8 column with measured numbers of the board: flash, arena, latency, and accuracy.
- ▶ ✓ The Decision Log gives numbers and explains the change of the accuracy.

Hand in: the file report.md. Show the kit to the instructor.

Write about 100 words. State one design decision, give your measured numbers, and name the trade-off.

The motion classifier of Day 3 goes into a product with the XIAO ESP32S3. Which model do you put on the board?

- ▶ The `float32` model.
- ▶ The `int8` model.

A good Decision Log:

- ▶ gives the flash, the arena, the latency, and the accuracy of the two models from your table,
- ▶ explains the change of the accuracy, or explains why the accuracy does not change. Use the clipped input values and your calibration set.
- ▶ names what you lose with your decision. Example: a smaller model against the work for the calibration set and the quantization code.

If a part does not work, and common problems

Problem	Fix or fallback
Your group has no dataset of Day 2	Copy the folder <code>data/</code> of a different group into <code>Labs/day02/</code> . Write this in the report.
Task 4 or task 5 is not complete in time	Copy the two functions from the notebook of <code>solutions/</code> . Parts C and D need them.
The <code>int8</code> model has an accuracy near 25 percent	The test code gives zeros or float values to the model. Check task 5.
Task 7: not complete, the gradient is <code>None</code>	The function returns <code>x_q</code> . Return <code>x + ops.stop_gradient(x_q - x)</code> .
The build prints <code>model_int8.h: No such file</code>	Run section 15 of the notebook, or use the files of the repository.
The board gives 20 correct windows of 80	The output is 0 for each class. Complete task D2.
same class as on the laptop is below 80	Check the scale and the zero point in tasks D1 and D2. Run section 15 again.
Tasks D1 and D2 are not complete in time	Use the sketch of <code>solutions/</code> with your four files of the notebook.

The README of the lab has the complete list.

This lecture adapts material from these sources:

- ▶ *Machine Learning Systems* by Vijay Janapa Reddi and contributors, Harvard University. mlsysbook.ai, CC BY-NC-SA 4.0.
- ▶ *Edge AI Engineering: Raspberry Pi* by Marcelo Rovai. github.com/Mjrovai/EdgeML_Made_Ease_ebook and github.com/Mjrovai/EdgeML-with-Raspberry-Pi, GPL-3.0.
- ▶ *TinyML Made Easy: XIAO ESP32S3* by Marcelo Rovai. github.com/Mjrovai/XIAO-ESP32S3-Sense, Apache-2.0.

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